

PRESENTATION 2026

TASKFLOW

PROJECT MANAGEMENT TOOL

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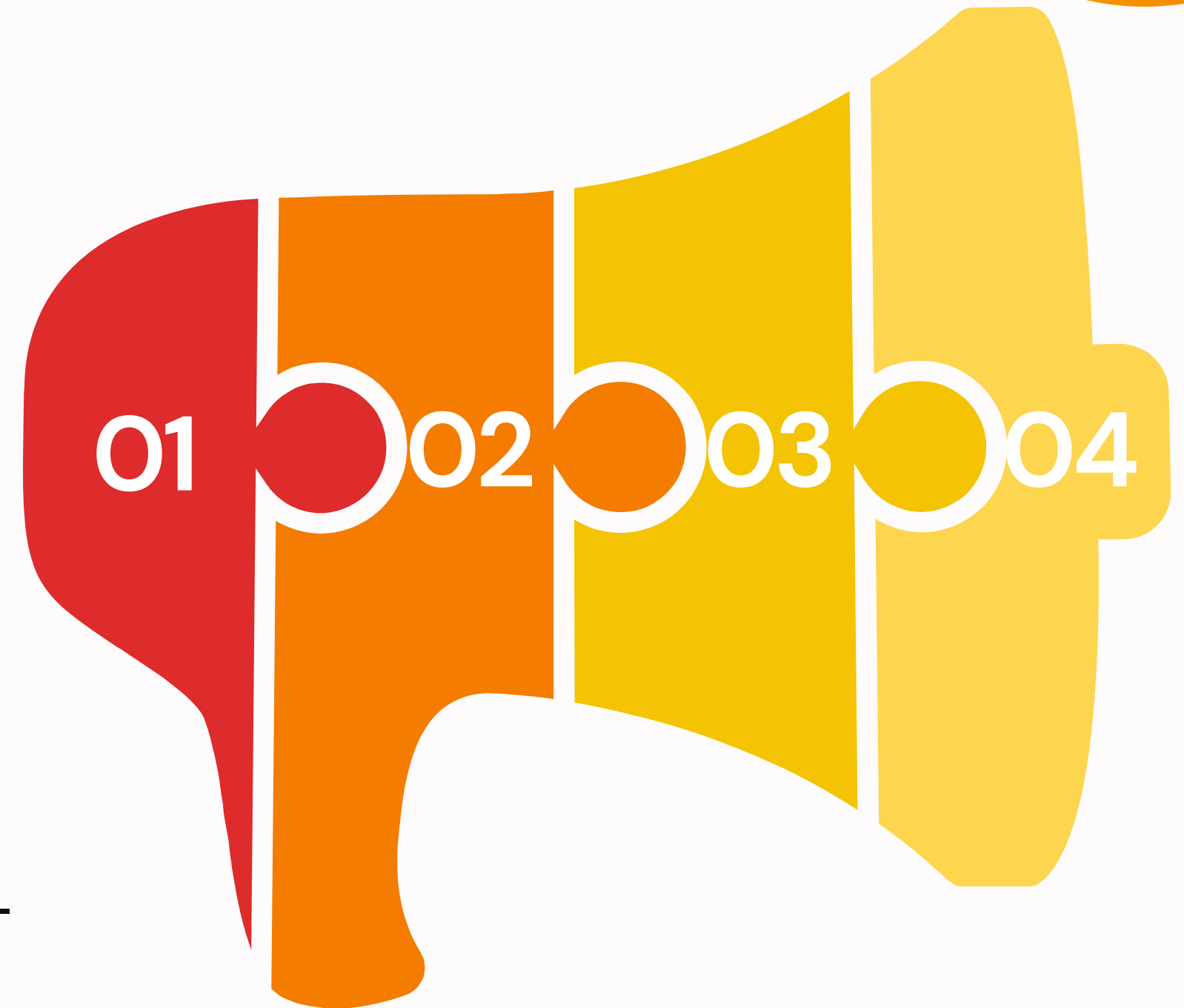
OVERVIEW

- 1. Project overview**
- 2. Users and Use cases**
- 3. Project Architecture**
- 4. My role and contribution**
- 5. Project Methodology**
- 6. Testing approach**
- 7. Future extensions**
- 8. Your experience/ lessons learnt.**

PROJECT OVERVIEW

The Kanban-Based Project Management System (TaskFlow) is a secure, role-based application that helps organizations manage projects and tasks efficiently

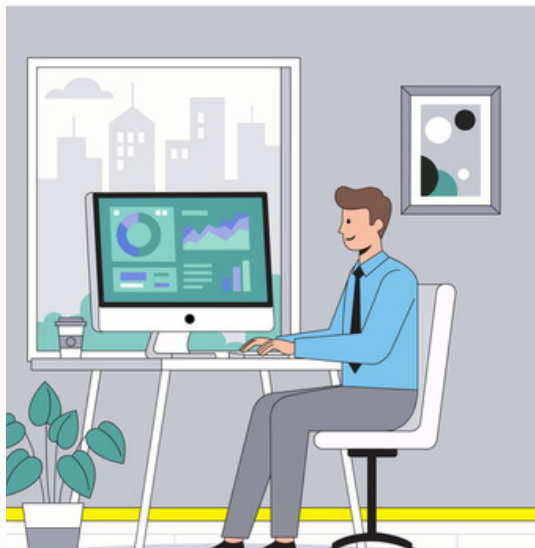
- 01** Role-based Kanban Project Management System for organizations
- 02** Supports Admin, Manager, Team Leader and Employee roles
- 03** Secure JWT authentication with domain-based access control
- 04** Responsive modern UI with real-time task tracking



ADMIN MODULE

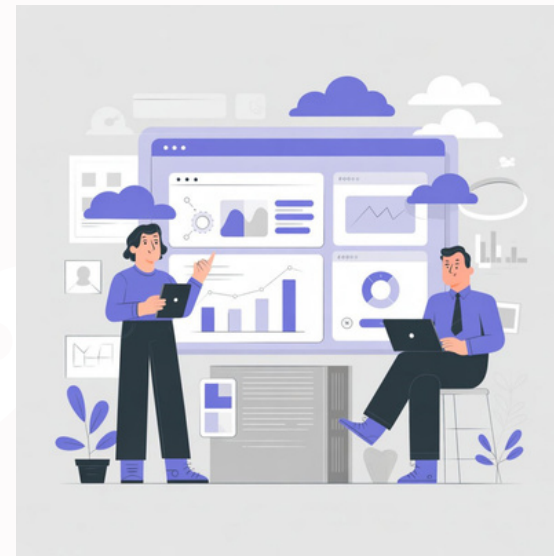
The Admin controls overall system access and **user management** by creating users, **assigning roles and domains**, and overseeing **project configuration**.

01



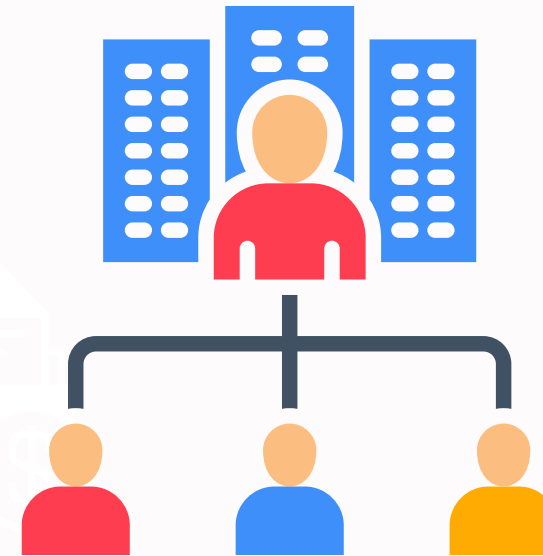
Manages user creation and role assignment

02



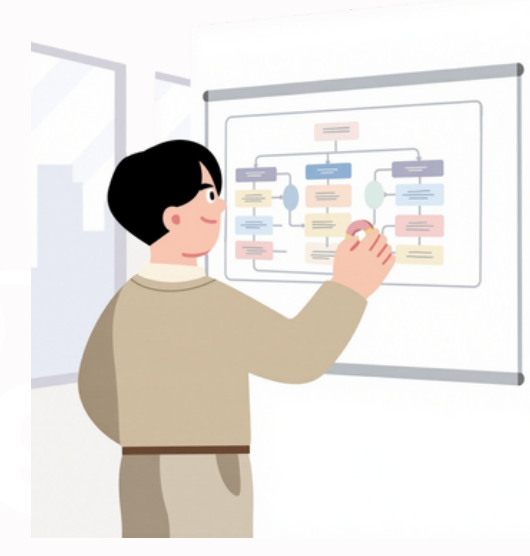
Controls domain-based access and permissions

03



Oversees project creation and allocation

04

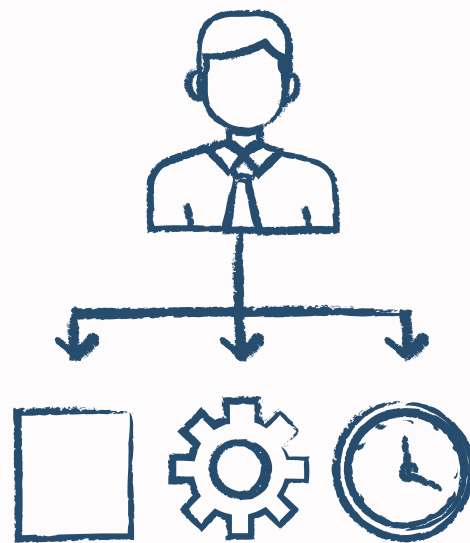


Monitors system activity through admin dashboard

MANAGER MODULE

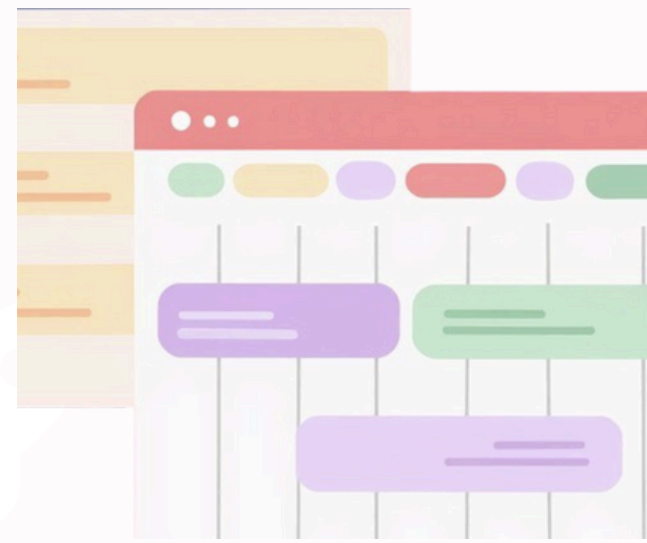
The Manager **oversees project execution** by managing projects assigned by the Admin and breaking them into domain-specific tasks. The **Manager assigns tasks** to Team Leaders, **tracks progress using a Kanban board**, and monitors overall project performance

01



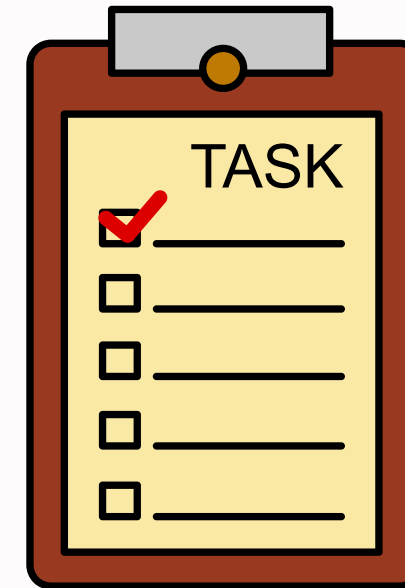
Manages projects assigned by the Admin

02



Tracks project progress using Kanban board

03



Breaks projects into domain-specific tasks

04



Assigns tasks to Team Leaders

TEAM LEADER MODULE

The Admin controls overall system access and **user management** by creating users, **assigning roles and domains**, and overseeing **project configuration**.

01



Coordinates and manages team-level tasks

02



Assigns subtasks to employees within the team

03



Tracks task progress through Kanban workflow

04



Reviews completed tasks and approves or rejects with feedback

EMPLOYEE MODULE

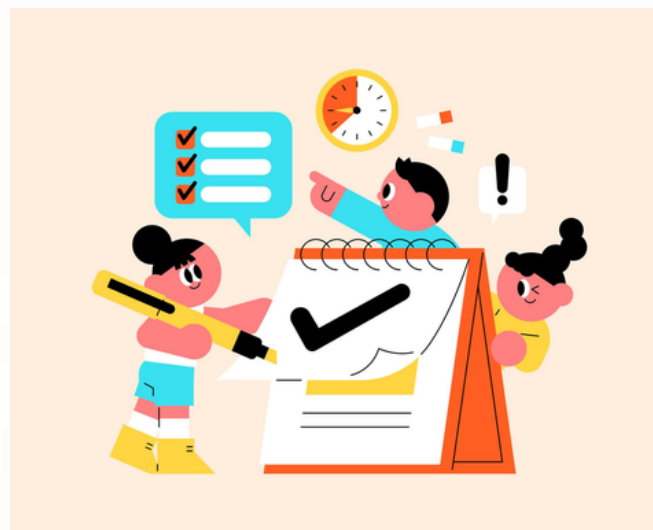
The Employee **works on tasks** assigned by the Team Leader and **updates task status** using the Kanban workflow. Employees **can raise queries** related to tasks and **receive feedback or comments** if a task is **rejected**. This ensures clear communication, accountability, and timely task completion.

01



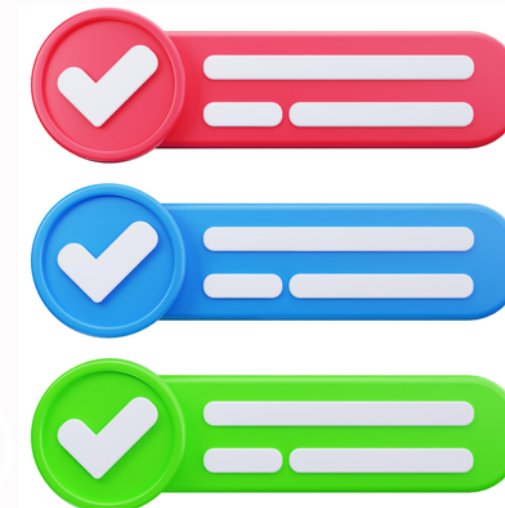
Views assigned tasks from Team Leader

02



Works on tasks within assigned domain

03



Updates task status (Pending, In Progress, Done)

04



Raises queries and receives feedback on tasks

Project Architecture

Frontend

Web server running on
localhost :3030

HTML,CSS

BOOTSTRAP

REACT JS Components

Middle tier with MICROSERVICES Architecture

(.NET Services)
Entity
Framework
ADO.NET

Discovery service
API Gateway

Business services

(Springboot
services)
JPA, Hibernate,
JDBC

Database

MySQL

01

02

Roles and Contribution

- Actively participated in requirement analysis to understand Task-flow's business workflow and project scope
- Contributed to system design and planning of a microservices-based architecture

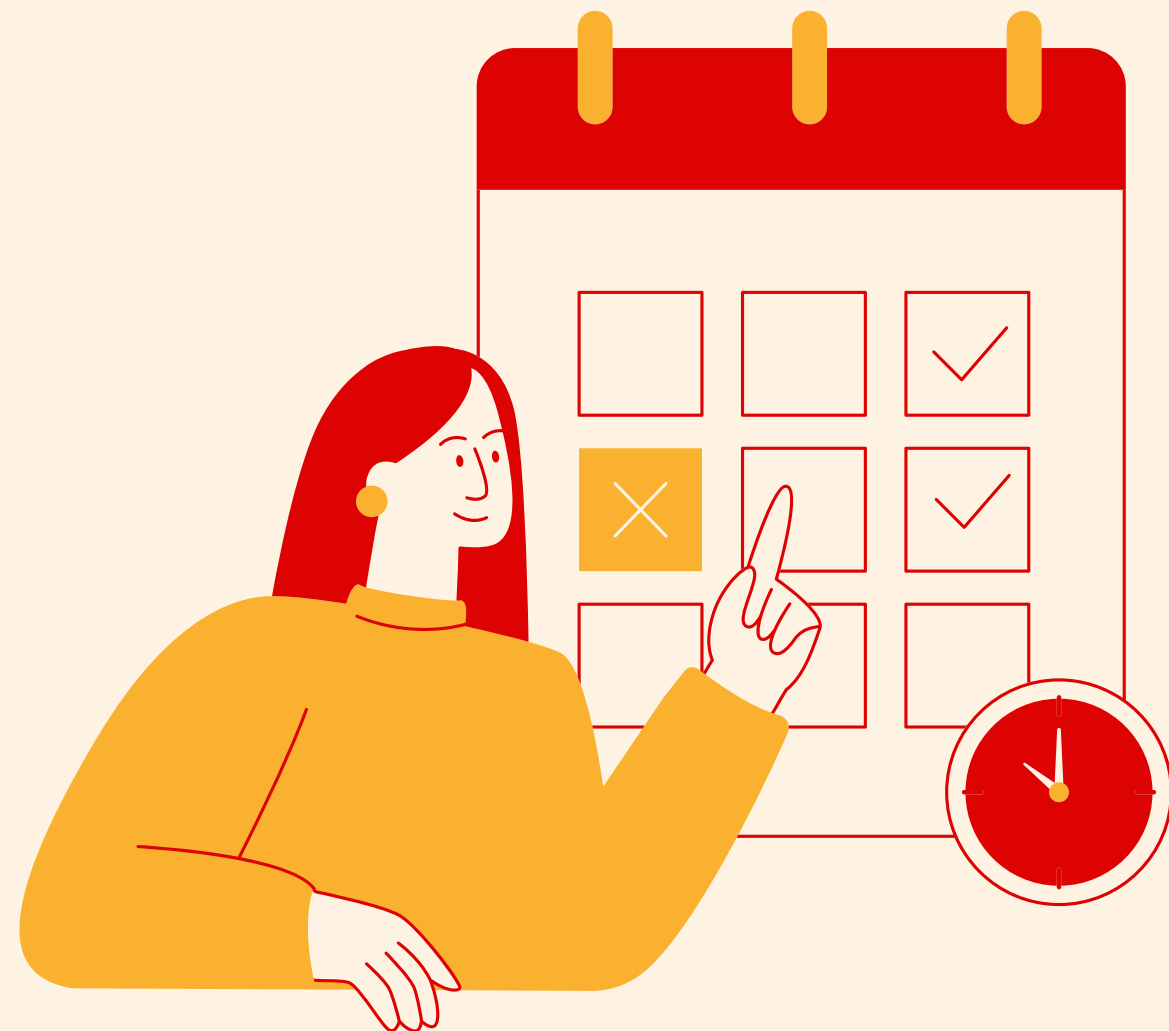
- Developed backend services using Spring Boot (Advanced Java) following a layered architecture
- Designed and implemented RESTful APIs for core Task-flow modules
- Worked on database design and integration using JPA/Hibernate

- Implemented key functionalities including user management, skill management, session scheduling, and booking workflows
- Contributed to points and transaction workflow integration to support Task-flow's business logic

- Assisted in frontend-backend integration using Axios for seamless API communication
- Performed unit testing, debugging, and bug fixes to ensure application stability
- Actively involved in project documentation and presentation preparation

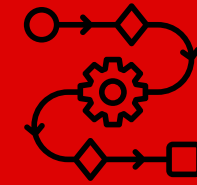


Project Methodology



Requirement Analysis

Gathered requirements based on project workflow and defined role-wise functionalities for Admin, Manager, Team Leader, and Employee.



System Design

Designed overall system architecture and Kanban-based task workflow with service-based modular structure.



Development & Integration

Developed backend services using Spring Boot and .NET, built frontend with React, and integrated using RESTful APIs.



Testing & Deployment

Performed iterative testing, fixed bugs, optimized performance, and prepared final deployment and documentation.

Testing Approach

Performed unit testing on individual modules and services
Tested backend APIs using Postman for request-response validation

Conducted functional testing to verify task workflows and role-based actions
Performed integration testing between Spring Boot and .NET services

Validated frontend components and user interactions
Tested Kanban board functionality for task movement and status updates

Conducted manual end-to-end testing across Admin, Manager, Team Leader, and Employee flows

Fixed identified bugs and performed regression testing to ensure stability



- Mobile application support for better accessibility across devices.
- Advanced reporting and analytics dashboards for Admins and Managers.
- Resource and workload optimization features for improved planning.
- Automated notifications and reminders for task deadlines and updates.
- Integration with third-party tools such as email and calendar systems.
- Enhanced security, access control, and performance optimization.



Future Extension

Experience / Lessons Learnt

- Gained practical experience in designing and implementing a role-based project management system.
- Developed hands-on understanding of service-based backend development using Spring Boot and .NET.
- Learned to design, consume, and integrate RESTful APIs between frontend and backend.
- Improved understanding of Kanban workflows and task lifecycle management.
- Gained experience in debugging, testing, and resolving integration issues.
- Improved coordination and collaboration through team-based development.
- Developed confidence in explaining system design, implementation, and project workflow.



Thank you